

Education

Bachelor of Technology – Electronics and Communication Engineering

CGPA: 7.71

Nov 2022 – May 2026

Senior Secondary (Class XII – CBSE)

Percentage: 91.4%

Secondary (Class X – CBSE)

Percentage: 89%

Coursework

- Data Structures & Algorithms
 - SQL
 - Object-Oriented Programming (OOP)
 - Database Management Systems
 - Operating Systems
 - Computer Networks
 - Web Development
 - Generative AI
 - Retrieval-Augmented Generation (Basics)
-

Projects

Full-Stack Real-Time Chat Application

Tech: React, Node.js, Socket.io, MongoDB

- Built a real-time chat application using the **MERN stack**, enabling message delivery with sub-150ms latency through WebSocket architecture.
 - Developed a scalable **RESTful API with 15+ endpoints** in Express.js and optimized MongoDB queries to maintain response times below 200ms.
 - Implemented **secure authentication using JWT**, significantly reducing unauthorized access risks and integrating security protection against brute-force attacks.
 - Integrated third-party APIs to enable **image sharing and automated welcome email notifications**.
-

Voice-Powered AI Virtual Assistant

Tech: React, Node.js, Express.js, MongoDB

- Developed a full-stack voice-controlled assistant using the MERN stack with **LLM-powered NLP capabilities**.
- Implemented real-time voice interaction using browser speech APIs to automate user tasks efficiently.

- Added secure token-based authentication and personalized user features with cloud storage integration for media handling.
-

Rubik's Cube Solver

Tech: C++, Data Structures, Algorithms

- Built a virtual **3×3 Rubik's Cube solver** using class-based representations and standard C++ data structures.
 - Implemented **BFS, DFS, and IDDFS algorithms** to solve cubes scrambled up to 8 moves within 3 seconds.
 - Integrated **IDA*** search algorithm to solve deeper scrambles up to 13 moves within 10 seconds.
-

Technical Skills

Languages

C, C++, JavaScript, SQL

Developer Tools

VS Code, Xcode, IntelliJ IDEA, Git

Technologies / Frameworks

HTML, CSS, Node.js, React.js, Express.js, MySQL, MongoDB

Extracurricular Activities

Competitive Programming

- Solved **600+ Data Structures and Algorithms problems** across competitive programming platforms.

Student Organization – Public Relations Lead

Aug 2023 – Present

- Led a team of five members to manage public relations and increase event participation through social media campaigns.
- Organized and promoted **15+ workshops** on public speaking, leadership, and professional development for over 500 students.